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Transverse dispersion of non-reactive tracers in porous media: A new nonlinear relationship to predict dispersion coefficients

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Abstract

Assessing the potential of natural attenuation in groundwater relies on the ability to predict and quantify the processes that occur in contaminant plumes. Transverse dispersion is a significant mass transfer mechanism for mixing of electron acceptors and donors and thus may control the lengths of steady state plumes.

Laboratory experiments were carried out using a 2-dimensional acrylic glass tank filled with glass beads, quartz sand and field site material as porous media. Flow velocities and grain sizes were varied in order to cover a large range of Peclet numbers including typical field scenarios. The laboratory study was extended by a comprehensive literature search to compare the new results with earlier work. As a result we propose a new empirical relationship for prediction of transverse dispersion coefficients (D_t) which is based on the Peclet number (Pe). This new relationship indicates a nonlinear dependency on the flow velocity (v_a) and grain size (d), namely a relative decrease of the dispersion coefficient with increasing flow velocity in relatively fast flowing water: $D_t/D_{aq}=D_p/D_{aq}+0.28(Pe)^{0.72}$ (with $Pe=v_a d/D_{aq}$; D_{aq} and D_p denote the aqueous and pore diffusion coefficients, resp.).
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1. Introduction

Since the early experimental work of Slichter (1905) a century ago, much effort has been invested into research of hydrodynamic dispersion in flow through porous media. Dispersion

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processes are attracting the attention of researchers within different branches of science and engineering. For example, chemical engineers are studying dispersion for applications such as chromatography (Baumeister et al., 1995) and chemical reactors (Balakotaiah and Chang, 1995). Petroleum engineers are investigating miscible-displacement processes in the recovery of oil (Blackwell, 1962). Medical science studies dispersion associated with transport processes in living organisms, for example red blood cells transported in the bloodstream (Bishop et al., 2002). Hydrologists are concerned with the risks of contamination of fresh-water supplies in coastal areas by the intrusion of salt water (De Josselin de Jong and van Duijn, 1986).

In groundwater research, and particularly in the field of contaminant hydrogeology, dispersion is currently studied in the context of transport and reactions of contaminants in aquifers. Specifically, for natural attenuation the understanding of dispersion processes is essential for reliable risk assessment. Natural attenuation in groundwater refers to the reduction in contaminant concentrations during migration from a source, e.g. a leaking underground storage tanks, in an aquifer. In groundwater, this reduction in concentration is due to a number of processes including dilution, dispersion, sorption, volatilisation, and biotic as well as abiotic transformations (Wiedemeier et al., 1999). The most important mass removal process for natural attenuation is biodegradation. Rates of biodegradation vary throughout a pollution plume as chemical concentrations, environmental conditions and microbiological activities vary. Transverse dispersion is believed to be the controlling mechanism for mixing electron donors in contaminant plumes with electron acceptors from the surrounding ground water (Cirpka et al., 1999; Cirpka and Kitanidis, 2000; Grathwohl et al., 2001; Ham et al., 2004). Mixing of reactants is a prerequisite for chemical reaction to take place and in effect, transverse dispersion is decisive for predicting lengths of steady state contaminant plumes (Grathwohl et al., 2000, 2001; Liedl et al., 2005; Cirpka et al., 2006; Maier and Grathwohl, 2006). Transverse dispersion is also the key process determining the rate of dissolution of non-aqueous phase liquids (especially pools) in aquifers (Eberhardt and Grathwohl, 2002) and transfer of volatile compounds from the unsaturated zone into the groundwater (Klenk and Grathwohl, 2002). This work aims to systematically elucidate the influence of grain size, grain size distributions, and variations in groundwater flow velocity on the magnitude of the transverse dispersion in saturated porous media.

2. Theory

A non-reactive tracer travelling through a saturated porous medium is subject to advection and dispersion. Dispersion causes the tracer to spread out over an area that is larger than it would be expected to occupy due to advection alone (Bear, 1972). Transverse dispersion causes spreading in the direction perpendicular to the direction of flow.

Our measurements of dispersion coefficients for a conservative tracer travelling through a porous medium utilises the solution to the advection–dispersion equation in 2D for a planar source (Domenico and Schwarz, 1990):

$$\frac{C}{C_0} = \frac{1}{2}\left\{ erf\left[\frac{(y+Y/2)}{2(D_t x/v_a)^{1/2}}\right] - erf\left[\frac{(y-Y/2)}{2(D_t x/v_a)^{1/2}}\right]\right\} \tag{1}$$

where D_t [m^2 s^{-1}] is the transverse hydrodynamic dispersion coefficient, C [mg m^{-3}] is the tracer concentration, C_0 is the concentration of the tracer at the source, Y [m] is the width of the source, and y [m] is the distance in the transverse direction. x/v_a [s] denotes the time the solute

needs to travel with flow velocity v_a [m s^{-1}] from the source to some observation point along the flow direction x [m]. D_t is traditionally defined as (e.g. Fetter, 1993):

$$D_t = D_p + \alpha_t v_a \tag{2}$$

where D_p [m^2 s^{-1}] is the pore diffusion coefficient and v_a the flow velocity. α_t [m] denotes the transverse dispersivity, which is representing a property of the porous medium. D_p can be estimated from following empirical relationship (Grathwohl, 1998; Boving and Grathwohl, 2001):

$$D_p = D_{aq} n \tag{3}$$

where D_{aq} [m^2 s^{-1}] is the aqueous diffusion coefficient and n [−] is the porosity. The product α_t v_a is termed hydromechanical dispersion coefficient. The use of Eq. (2) to describe dispersion assumes that at high values of α_t v_a (or relatively small values of D_p), D_t linearly increases with the flow velocity. Then the shape of concentration profile (e.g. in Eq. (1)) only depends on the distance travelled and α_t, i.e. the concentration profile is independent on the flow velocity.

Several investigations of the factors that influence the magnitude of the transverse dispersivity have already been carried out. According to classical theory developed by De Josselin de Jong (1958) the relation between flow velocity, grain size and hydrodynamic transverse dispersion coefficient equals:

$$D_t = D_p + \frac{3}{16} v_a d \tag{4}$$

where d [m] denotes the effective grain size or grain diameter. If data for different grain sizes and flow velocities have to be compared, often the dimensionless Peclet number is employed:

$$Pe = \frac{v_a d}{D_{aq}} \tag{5}$$

Using measurements of the rate of dissolution of a soluble cylinder embedded in a porous medium, which also depends on transverse mixing, Carvalho and Delgado (1999) found that for $Pe<80$, Eq. (4) has to be modified:

$$D_t = D_p + \frac{1}{12} v_a d \tag{6}$$

Furthermore, according to Carvalho and Delgado (2000), Eq. (6) becomes invalid in the range $80<Pe<1400$ and the dispersion coefficient is better represented by:

$$D_t = D_p + D_{aq} 0.50 (Pe)^{0.57} \tag{7}$$

Similarly, Klenk and Grathwohl (2002), who measured mass fluxes of volatile compounds across the capillary fringe in a bench scale tank experiment, found a nonlinear relationship between D_t and v_a. They concluded that in a coarse sand above a critical flow velocity of around 4 m day^{-1} transverse dispersion is non-linearly related to the flow velocity and derived following relationship based on theoretical considerations on incomplete mixing in pore throats:

$$D_t = D_p + D_{aq} \frac{3}{2} \sqrt{\frac{4}{\pi} Pe} \tag{8}$$

According to Eqs. (7) and (8), D_t increases non-linearly with increasing flow velocity in contrast to Eqs. (2), (4) and (6) (if D_p can be neglected). This means that α_t would apparently decrease with increasing flow velocity if Eq. (2) is used to derive α_t from measurements of D_t in the high velocity region as shown by Klenk and Grathwohl (2002). Results from Huang (2001) also indicate a decrease in dispersivity values above a flow velocity of approximately 31 m day^{-1}.

Other literature data, mainly from the area of chemical engineering, where typically high flow velocity regimes are studied, clearly indicate that the established hydrogeological theory postulating that the transverse dispersion coefficient is linearly related to flow velocity (e.g. Eq. (2)) does not hold. In this study, we aim to bridge the gap between slow groundwater flow and the high velocities used in chemical engineering. Therefore transverse dispersion coefficients were measured over a wide velocity range in aquifer materials and glass bead porous media in order to elucidate the relation between flow velocity, grain size and transverse dispersion coefficients.

3. Materials and methods

The tracer experiments were carried out in two quasi-2D acrylic-glass flow-through tanks, a smaller one of dimensions ($L \times H \times W$) 28.0 cm × 14.0 cm × 1.05 cm and a larger one of 77.9 cm × 14.0 cm × 0.8 cm. The different tank sizes were used to investigate whether the measurements were subject to any scale effects. Both tanks are equipped with 11 ports each at the inlet and outlet ends. Parallel and steady flow through the vertical tanks was maintained by employing two 12-channel peristaltic pumps operated at the same rate, one at the inlet and one at the outlet side, as shown in Fig. 1.

The pumps were connected to the tank through tygon tubing with 0.57 mm inner diameter, and stainless steel tubing with an inner diameter 0.75 mm. Granular medium was filled into the tank using a funnel. In order to achieve a uniform packing without trapping air, the water level was always kept above the grains during filling. Deionized water was injected through 10 inlet ports, spaced at 1.1 cm intervals, while a conservative tracer was injected through the middle port (the 12th channel of the pump was not used). The conservative tracer solution was fluorescein at a concentration of 5 mg l^{-1}. For comparison a second tracer, bromide, was used in a concentration of 20 mg L^{-1} in some experiments. To make sure that the flow through the tank was stable and uniform, the flow through individual pump tubings was measured regularly and the settings were adjusted accordingly. The difference in flow rates between the different lines was less than 10%. Preliminary tests showed, that for stable laminar flow two pumps, one for the inlet and one for the outlet, had to be used. When the plume was at steady state, samples were collected at the outlet

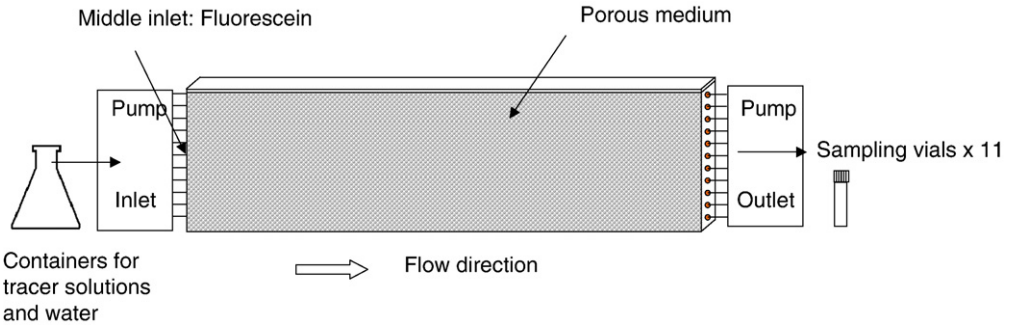

Fig. 1. Schematic drawing of the experimental setup.

Table 1
Conditions and parameters obtained in the tracer experiments

v_a (m d^{-1})	Pe ($v_a d/D_{aq}$)	D_t (m^2 s^{-1})	α_t (mm)	Tank length (cm)	Porous medium grain-size
	Fluorescein [a]				0.10–0.25 mm
0.37	1.2	5.0×10^{-10}	6.3×10^{-2}	77.9	Glass beads
0.68	2.1	6.9×10^{-10}	5.8×10^{-2}	77.9	”
0.88	2.7	6.2×10^{-10}	3.8×10^{-2}	28.0	”
1.0	3.1	6.8×10^{-10}	3.8×10^{-2}	77.9	Quartz
1.4	4.5	7.7×10^{-10}	3.2×10^{-2}	77.9	Glass beads
1.4	4.3	7.7×10^{-10}	3.4×10^{-2}	77.9	Quartz
1.5	4.7	8.0×10^{-10}	3.2×10^{-2}	28.0	Glass beads
4.1	12.9	1.1×10^{-9}	1.8×10^{-2}	77.9	”
4.6	14.5	1.3×10^{-9}	2.0×10^{-2}	28.0	”
4.6	14.2	1.5×10^{-9}	2.4×10^{-2}	77.9	Quartz
7.4	23.2	1.6×10^{-9}	1.6×10^{-2}	28.0	Glass beads
13.8	43.3	3.1×10^{-9}	1.8×10^{-2}	28.0	”
					0.10–0.25 mm, 4 °C [b]
0.38	1.2	2.9×10^{-10}	3.7×10^{-2}	28.0	Glass beads
0.41	1.3	2.4×10^{-10}	1.9×10^{-2}	28.0	”
1.0	11.3	5.5×10^{-10}	4.2×10^{-2}	28.0	”
1.3	4.2	3.4×10^{-10}	1.9×10^{-2}	28.0	”
1.7	5.4	6.3×10^{-10}	2.9×10^{-2}	28.0	”
2.2	6.9	5.5×10^{-10}	2.0×10^{-2}	28.0	”
5.2	16.4	1.4×10^{-9}	2.2×10^{-2}	28.0	”
					0.50–0.75 mm
0.68	7.6	1.1×10^{-9}	1.1×10^{-1}	77.9	Glass beads
2.3	25.7	2.4×10^{-9}	8.1×10^{-2}	77.9	”
4.0	44.3	2.7×10^{-9}	5.3×10^{-2}	77.9	”
22.3	249	8.8×10^{-9}	3.3×10^{-2}	77.9	”
92	1027	3.5×10^{-8}	3.3×10^{-2}	77.9	”
154	1714	5.1×10^{-8}	2.9×10^{-2}	77.9	”
					1.0–1.5 mm
0.6	13.0	1.6×10^{-9}	2.0×10^{-1}	77.9	Glass beads
1.8	39.7	2.5×10^{-9}	1.1×10^{-1}	77.9	”
1.8	40.6	2.5×10^{-9}	1.1×10^{-1}	77.9	”
1.9	43.1	2.4×10^{-9}	9.6×10^{-2}	28.0	”
3.4	76.0	2.9×10^{-9}	6.7×10^{-2}	28.0	”
3.0	67.3	3.0×10^{-9}	7.9×10^{-2}	77.9	”
17.3	385	1.1×10^{-8}	5.3×10^{-2}	77.9	”
17.4	389	1.1×10^{-8}	5.3×10^{-2}	77.9	”
17.4	388	1.1×10^{-8}	5.3×10^{-2}	77.9	”
18.0	401	1.4×10^{-8}	6.6×10^{-2}	28.0	”
64.4	1437	3.8×10^{-8}	5.1×10^{-2}	77.9	”
69.3	1547	3.5×10^{-8}	4.3×10^{-2}	77.9	”
71.4	1594	4.5×10^{-8}	5.4×10^{-2}	28.0	”
71.5	1596	3.8×10^{-8}	4.6×10^{-2}	77.9	”
117.6	2625	5.7×10^{-8}	4.1×10^{-2}	77.9	”
118.3	2642	5.5×10^{-8}	4.0×10^{-2}	77.9	”
119.4	2665	5.5×10^{-8}	3.9×10^{-2}	77.9	”
124.0	2775	5.5×10^{-8}	3.8×10^{-2}	28.0	”
					2.0–2.3 mm
0.7	26.9	2.7×10^{-9}	3.0×10^{-1}	77.9	Glass beads
2.0	68.5	3.2×10^{-9}	1.4×10^{-1}	77.9	”
4.3	166	6.2×10^{-9}	1.2×10^{-1}	77.9	”

(continued on next page)

Table 1 (*continued*)

v_a (m d^{-1})	*Pe* ($v_a d/D_{aq}$)	D_t (m^2 s^{-1})	α_t (mm)	Tank length (cm)	Porous medium grain-size
	Fluorescein [a]				2.0–2.3 mm
22.3	855	2.0×10^{-8}	7.7×10^{-2}	77.9	”
95.6	3671	6.0×10^{-8}	5.4×10^{-2}	77.9	”
164	6307	1.2×10^{-7}	6.3×10^{-2}	77.9	”
	Bromide [c]				0.10–0.25 mm
0.18	0.2	1.20E-09	0.224	28.0	Glass beads
0.54	0.6	2.00E-09	0.202	28.0	”
1.9	1.9	4.50E-09	0.121	28.0	”
3.1	3.2	5.00E-09	0.088	28.0	”
18.7	18.9	1.56E-08	0.066	28.0	”
	Non-uniform materials (grain size mixtures)				
	Field site aquifer material: Polimeri Europa I				$d(U=d_{60}/d_{10})$[d]
0.58	4.0	6.0×10^{-10}	5.4×10^{-2}	28.0	0.38 (2.2)
2.2	14.8	1.8×10^{-9}	6.2×10^{-2}	28.0	”
7.8	52.9	1.7×10^{-9}	1.6×10^{-2}	28.0	”
	Field site aquifer material: Polimeri Europa II				$d(U=d_{60}/d_{10})$[d]
0.48	2.8	4.5×10^{-10}	3.5×10^{-2}	28.0	0.33 (1.5)
2.0	11.9	1.3×10^{-9}	4.5×10^{-2}	28.0	”
4.0	23.9	3.6×10^{-9}	7.1×10^{-2}	28.0	”
8.6	50.9	4.2×10^{-9}	3.9×10^{-2}	28.0	”
	Field site aquifer material: Borden Test Site				$d(U\ (=d_{60}/d_{10})$[d]
0.68	2.3	4.0×10^{-10}	2.1×10^{-2}	28.0	0.19 (1.6)
1.2	4.2	5.0×10^{-10}	1.9×10^{-2}	28.0	”
9	30.5	1.7×10^{-9}	1.4×10^{-2}	28.0	”

All data for 25 °C unless otherwise noted. The porosity of all media was between 0.34 and 0.36, for calculation of v_a and D_p a value of 0.35 was used. *Pe* was calculated based on the arithmetic mean of the grain sizes.

[a] $D_{aq}=6.48\times10^{-10}$ m^{2}s^{-1}.

[b] $D_{aq}=3.57\times10^{-10}$ m^{2}s^{-1}.

[c] $D_{aq}=2.08\times10^{-9}$ m^{2}s^{-1}.

[d] *U*: non-uniformity index.

end and diluted up to 10 000 times, depending on the concentration. The fluorescein concentration of each sample was analysed using a Perkin Elmer LS-3B fluorescence spectrometer.

The experiments were conducted with porous media consisting of glass beads and quartz sand, both of uniform grain diameters. Silica glass beads of ø 0.10–0.25, 0.50–0–75, 1.0–1.5 and 2.0–2.3 mm and a quartz sand of ø 0.10–0.25 mm were used. In addition to the media consisting of a single grain size, eight different mixed-grain-size porous media consisting of several grain sizes in different proportions were prepared using glass beads. Furthermore, three field site aquifer materials from natural attenuation field study sites were used: one sample from the Borden test site in Canada, and two from the Polimeri Europa field site in Mantova, Italy. For each porous medium, the tracer distribution was measured for a series of flow velocities ranging between 0.4 and 160 m day^{-1}. After each experiment the tank was emptied, the next granular material washed and the tank refilled. Before starting a new experiment, the porous medium was flushed with at least one pore volume of deionized water before introducing the tracer.

A number of tests were performed to prove the robustness of the setup and to check for experimental artefacts (see Table 1). To elucidate the influence of temperature, one set of experiments was performed at a constant temperature of 4 °C and the results were compared to the hydrodynamic dispersion at room temperature. As expected, the different temperatures had no effect on the dispersivities determined. The reproducibility of the method was investigated by

disassembling and reassembling the experimental setup and repeating some of the experiments from scratch. This provided a means of investigating how well the homogeneity of the packing and the stability of the flow agree between experiments. No significant difference in data was measured for the short and the long tanks, which indicates that the measurements are not sensitive to scale when the porous medium is homogeneously packed. Using tanks of two different lengths additionally proves that inlet conditions do not influence the tracer distribution. To investigate whether the trends observed in the data are due to some property inherent of the fluorescein tracer, experiments were repeated under similar experimental conditions using NaBr (sodium bromide). Dispersion of bromide ions was studied using the same setup as above, using glass beads of ø 0.10–0.25 mm, and five flow velocities. Bromide concentrations were analysed using Merck-Hitachi HPLC F-1080 fluorescence detector.

4. Data analysis

The fluorescence signal for each sample relative to the signal of the injected solution (C/C_o) was plotted against the lateral distance from the injection port (y). Eq. (1) was fitted to the plotted data points, using D_t as the fitting factor (see example in Fig. 2). In some cases, where flow was not exactly parallel, y had to be shifted slightly to match the peak concentration at the outlet. The dispersion coefficient was adjusted until the analytical solution fits the data set as closely as possible by minimising the sum of the squared error between the measured data and Eq. (1). From the fitted D_t-values α_t was calculated based on Eq. (2). D_{aq} values used for fluorescein and bromide were 6.48×10^{-10} m^2s^{-1} and 2.08×10^{-9} m^2s^{-1}, respectively (Ju, 2000; Willingham et al., 2004).

5. Results and discussion

The results of the tracer experiments show that transverse dispersivities measured at laboratory scale in homogeneous, saturated porous media are very small even for coarse sands. All transverse dispersivities measured here lie between 18 μm and 300 μm. Most literature values obtained for similar conditions (i.e. homogeneous, saturated porous media at local scale) are of a

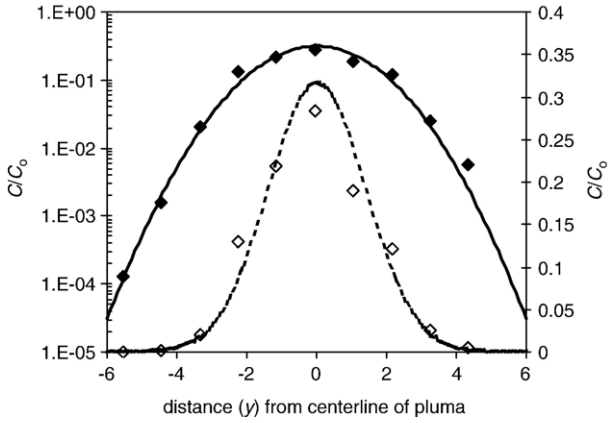

Fig. 2. Examples for fitting Eq. (1) to experimental concentration profiles to determine D_t. Solid line and filled symbols: left scale (logarithmic); open symbols, dashed line: right scale (linear).

similar magnitude, as summarised e.g. in Huang (2001) or Klenk and Grathwohl (2002). Tables 1 and 2 list the results of the fluorescein and bromide tracer experiments from this study.

In Fig. 3 α_t values obtained at four different flow velocities are plotted versus the grain size. Additionally, data from Grane and Gardner (1961) were included and compared to predictions from Eqs. (4) and (8). In general, transverse dispersivities increase with increasing grain diameter. Such a positive correlation between transverse dispersivity and grain diameter has previously been observed (e.g. De Josselin de Jong, 1958; Grane and Gardner, 1961). Ju (2000) found in laboratory studies that α_t is 10% of the grain diameter. For our data, this increase is smaller than the ratio proposed in the classical work by De Josselin de Jong (1958), whose theoretically derived equation predicts a linear relation between grain size and transverse dispersivity as described by Eq. (4). This relation fits relatively well to experimental data for the slowest flow velocity but it deviates more and more from our measurements at higher flow velocities. Fig. 3 shows how experimental data are increasingly well predicted by the equation proposed by Klenk and Grathwohl (2002) (Eq. (8)), and for the highest flow velocity the fit is excellent.

Fig. 3 also shows that the relationship between grain diameter and transverse dispersivity is not the same for each flow velocity. At the lowest flow velocity (0.5 m day^{-1}), $\alpha_t = 3/16d$ and at the highest flow velocity (120 m day^{-1}) it is only $1/70d$. Apparently the importance of grain size is less pronounced at higher flow velocity.

Fig. 4 compiles all values of α_t calculated with Eq. (2) from D_t measured in this study. The result clearly shows that transverse dispersivity apparently decreases with increasing flow velocity and it should be noted, that this is already pronounced in the region of typical groundwater flow velocities (0.1–10 m day^{-1}).

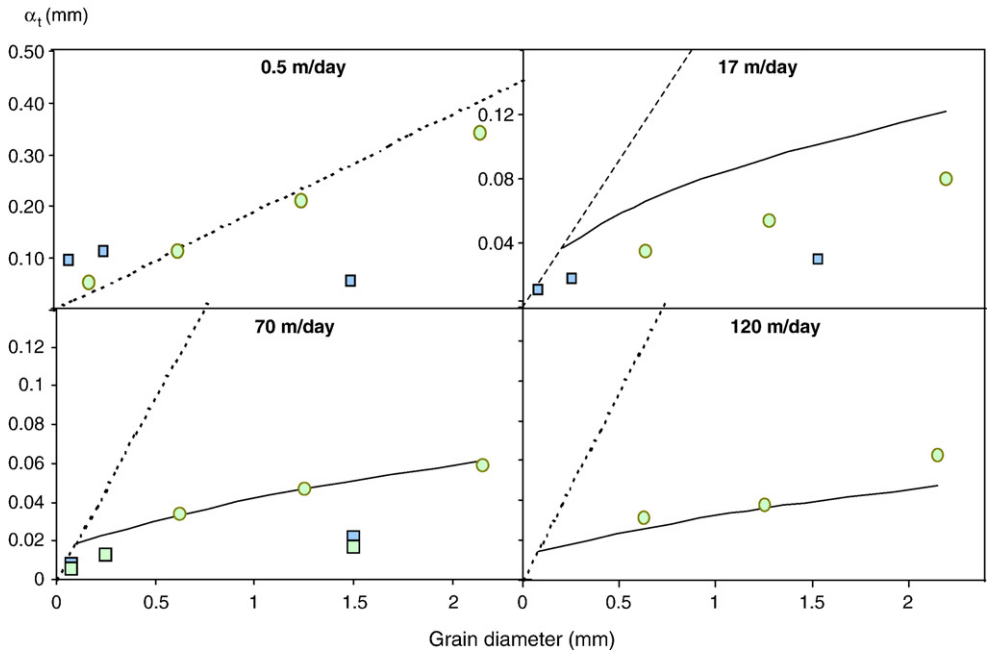

Fig. 3. Relation between transverse dispersivity (α_t) and grain diameter for different flow velocities. Circles: this study; squares: Grane and Gardner (1961); dashed line: Eq. (4) (De Josselin de Jong, 1958); solid line: Eq. (8) (Klenk and Grathwohl, 2002).

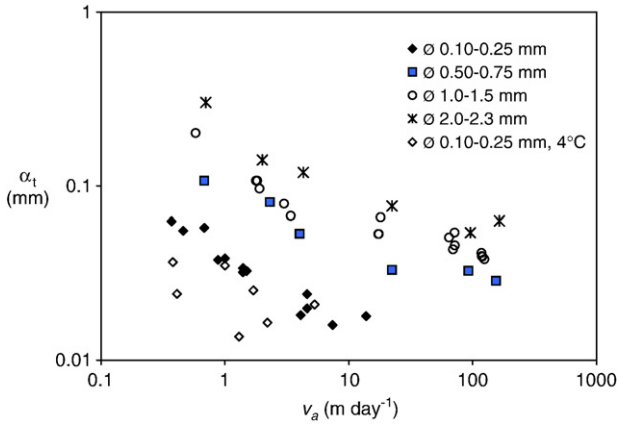

Fig. 4. Apparent relation between transverse dispersivity (α_t) and flow velocity if Eq. (2) is applied to calculate α_t. Data are for fluorescein tracer in short and long tanks including quartz sand and glass beads and data from experiments carried out at 4 °C.

From Eqs. (7) and (8) we expect that a general nonlinear equation to predict D_t should have following general form:

$$D_t = D_p + D_{aq} b (Pe)^m \tag{9}$$

where b and m are empirical constants. All data obtained from the fluorescein experiments with the uniform grain sizes (Table 1) were evaluated following Eq. (9) in a linearized, i.e. logarithmic form (see Fig. 5):

$$\log\left(\frac{D_t}{D_{aq}} - \frac{D_p}{D_{aq}}\right) = \log\, b + m\, \log\, Pe \tag{10}$$

The slope m and intercept log b were determined by linear regression analysis as 0.72 and −0.55 with standard deviations of 0.012 and 0.027, respectively. The average absolute normalized error

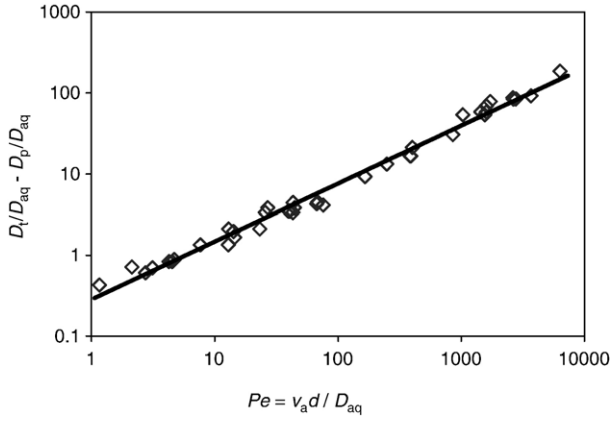

Fig. 5. Linear regression Eq. (10) for prediction of D_t from Peclet numbers (Pe), data for uniform grain sizes (Table 1).

between predicted and measured values was 13%, R^2 was 0.99. From this analysis we propose a new empirical equation where D_t is predicted as a non-linear function of Pe:

$$\frac{D_t}{D_{aq}} = \frac{D_p}{D_{aq}} + 0.28(Pe)^{0.72} \tag{11}$$

In Fig. 6, results from this study and of other experimental studies are compared to Eq. (11) as well as other literature relationships. The plot includes experimental data using porous media consisting of one single grain size, mixed porous media, field site materials, long and short tanks, glass beads and quartz sand. Results from bromide tracer experiments are also plotted in the same graph.

5.1. Application of the new empirical relationship to mixed grain sizes

In porous media with a non-uniform grain size the question arises whether the arithmetic or geometric means should be used to estimate D_t using Eq. (11). To test this, in total 9 mixtures of the glass bead sample listed in Table 1 were employed in additional experiments: 5 mixtures consisted of 25% of each grain size, 33% of the three smallest grain sizes, 33% of the three largest grain sizes and two mixtures of 50% of the two smallest and 50% of the two largest grain sizes. 4 mixtures consisting of 20% of 2.0–2.3 mm in 0.50–0.75 mm and vice versa, 20% of 2.0–2.3 mm in 1.0–1.5 mm and 20% of 1.0–1.5 mm in 0.50–0.75 mm. Fig. 7 shows the plot of the transverse dispersion coefficients predicted using Eq. (11), against measured values for the porous media consisting of mixed glass beads and the field materials (see Table 1) using the arithmetic means of the grain sizes. The average absolute error between measured and predicted values is 31% with no significant trend towards over- or underprediction. These results indicate that the transverse dispersion coefficients for mixed size porous media are also well predicted by the new relationship. This conclusion is also supported by Carvalho and Delgado (2000), who observed that dispersion in a bed with a mixture of particle sizes follows the prediction for beds of nearly mono-sized particles if the arithmetic mean for particle size

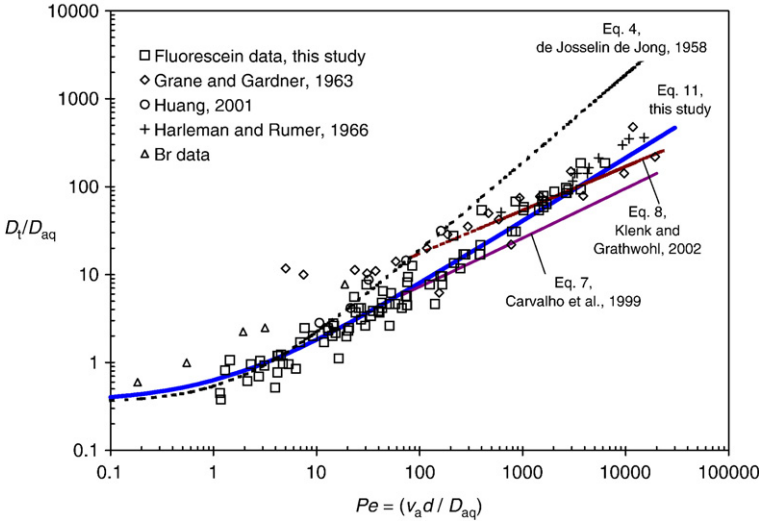

Fig. 6. Transverse dispersion coefficients form this and other studies plotted against the Peclet number and compared to empirical equations for prediction of D_t (Harleman and Rumer, 1963).

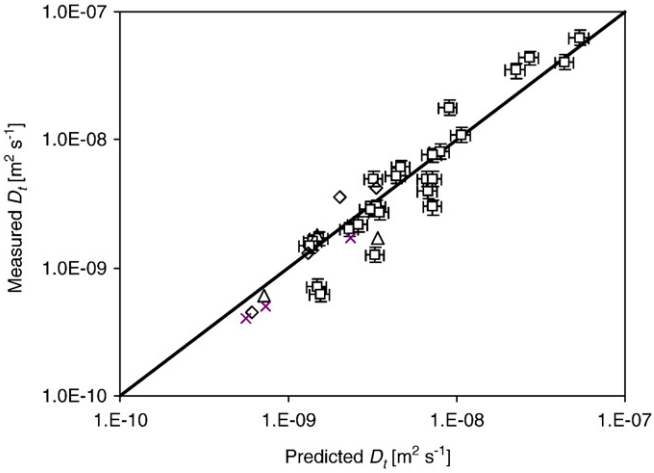

Fig. 7. Transverse dispersion coefficients (m^2/s) predicted for mixed grain sizes of glass beads (squares) and field site materials (Table 1, Polymeri I: triangles, Polymeri II: diamonds; Borden: x) using Eq. (11) based on the arithmetic means of the grain sizes (error indicators: 13% average absolute error).

is used. Additionally, for the materials composed of mixed size glass beads and for the field site materials, the uniformity index $U(d_{60}/d_{10})$ was calculated. The uniformity index is the ratio of the grain diameters of 60% to the grain diameters of 10% of the samples. Again, no significant correlation between sorting of the grains and transverse dispersion coefficient was observed.

6. Conclusions

To our knowledge, the experimental data presented here cover a larger range of Peclet numbers than has hitherto been investigated in one and the same experimental study. Local values of transverse dispersivity in porous media are very small even in coarse sands, as shown by results from the current and other studies. All values for α_t are below 1 mm in a range from 18 μm to 300 μm and correspond well to values reported in literature. The tank method developed for the determination of local-scale transverse dispersion coefficients is robust and reliable. Repeated experiments deliver reproducible values also after emptying and refilling the tank, which shows that the medium was packed homogeneously and that the error in the measurements is reasonably small. Cross-checks using three different types of porous media in two different sized tanks, two different tracers and performing experiments at 4 °C as well as at room temperature prove that the experimental method is robust and yields reproducible results.

The results support a new empirical relationship for predicting pore scale transverse dispersion coefficients. This equation describes dispersion as a function of the Peclet number and thus accounts for grain size, flow velocity and aqueous diffusion coefficient of the migrating substance. It is valid for the entire range of Peclet numbers investigated in this study. According to this new relationship, hydrodynamic dispersion is always influenced by molecular diffusion. For $Pe < 10$ diffusion contributes significantly to transverse dispersion. For Pe between 10 and 70 the experimental dispersion coefficients shows an apparently linear increase with the flow velocity, but the data scatter is high. At higher Peclet numbers (>70), D_t is increasing at a slower rate

presumably because slow molecular diffusion does not allow for complete mixing in pore throats. For mixed granular porous media, the arithmetic mean of the grain size seems to be appropriate to predict D_t with Eq. (11).

If these laboratory data with very small pore scale transverse dispersivities are used to predict the lengths of steady state contaminant plumes in real aquifers, then the result will be very narrow and extremely long plumes. For up-scaling of the dispersion parameters, it will be necessary to quantify additionally the influence of heterogeneities (e.g. sedimentary structures; Werth et al., 2006), sorption (including sorption/desorption kinetics) and transient flow conditions on transverse mixing (Cirpka and Attinger, 2003). Each of these effects can increase transverse mixing at the field scale significantly (roughly by half an order to one order of magnitude). The overall magnitude of these increases has to be investigated in further experiments and comprehensive numerical simulations.

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